

JAVA TYPE CASTING

Complete Trainer + Student Master Notes

Concepts • Examples • Traps • Practice • Interview Questions

1. What is Type Casting?

- Type casting means converting a value from one data type to another compatible data type.

```
int x = 100;
double y = x;
System.out.println(y);

double price = 99.99;
int p = (int) price;
System.out.println(p);
```

2. Widening Casting — Automatic

- Small-range numeric type → larger-range numeric type. Java usually performs this automatically.

```
byte b = 10;
int i = b;
long l = i;
double d = l;
System.out.println(d);

int marks = 90;
double percentage = marks;
System.out.println(percentage);
```

3. Narrowing Casting — Manual

- Larger numeric type → smaller numeric type generally needs an explicit cast. Information may be lost.

```
double d = 99.99;
int x = (int)d;
System.out.println(x);

int x = 130;
byte b = (byte)x;
System.out.println(b);
```

4. Decimal Truncation Trap

double/float → integer discards the fractional part; it does not round.

```
double x = 10.9;
System.out.println((int)x); // 10
System.out.println(Math.round(x)); // 11
```

5. Integer Division Trap

- Both operands int means integer division happens before any cast placed around the complete expression.

```
int a = 5;
int b = 2;
double x = (double)(a / b);
double y = (double)a / b;
System.out.println(x);
System.out.println(y);
```

6. Primitive Promotion in Arithmetic

- byte, short and char are commonly promoted to int during arithmetic operations.

```
byte a = 10;
byte b = 20;
int c = a + b;
System.out.println(c);

char ch = 'A';
int n = ch + 1;
System.out.println(n);
```

7. Compound Assignment Special Case

- Compound assignments such as += include an implicit conversion to the left-hand type.

```
byte b = 10;
b += 5;
System.out.println(b);

byte b = 10;
// b = b + 5; // compile-time error
b += 5; // valid
```

8. char and Numeric Casting

- char stores a UTF-16 code unit and can participate in numeric conversions.

```
char ch = 'A';
int n = ch;
System.out.println(n);

int n = 66;
char ch = (char)n;
System.out.println(ch);
```

9. byte Overflow / Narrowing

- Narrowing an integer can keep only the target type's low-order bits, producing a surprising result.

```
int x = 130;
byte b = (byte)x;
System.out.println(b);

int x = 257;
byte b = (byte)x;
System.out.println(b);
```

10. Floating-Point Precision

- Range widening does not guarantee mathematically exact decimal representation because float/double use binary floating-point.

```
float f = 10.5f;
double d = f;
System.out.println(d);

double d = 10.5;
float f = (float)d;
System.out.println(f);
```

11. Reference Casting — Basic

Primitive casting and object/reference casting are different. Upcasting is generally automatic; downcasting needs an explicit cast.

```
class Animal { }
class Dog extends Animal { }

Animal a = new Dog();
Dog d = (Dog)a;
```

- Use instanceof before a potentially unsafe downcast when appropriate.

```
if (a instanceof Dog) {  
    Dog d = (Dog)a;  
}
```

12. Common Mistakes

- $(\text{double})(5/2) \rightarrow 2.0$ because integer division happens first.
- $(\text{double})5/2 \rightarrow 2.5$ because one operand becomes double before division.
 $(\text{int})9.99 \rightarrow 9$; casting does not round.
- $\text{byte} + \text{byte} \rightarrow \text{int}$ in normal arithmetic.
- boolean cannot be numerically cast to int.
- Reference downcasting can fail with `ClassCastException`.

13. Practice — Predict the Output

Q1.

```
int x = 10;  
double y = x;  
System.out.println(y);
```

Output: _____

Q2.

```
double x = 9.99;  
int y = (int)x;  
System.out.println(y);
```

Output: _____

Q3.

```
int a = 5, b = 2;  
System.out.println((double)(a / b));
```

Output: _____

Q4.

```
int a = 5, b = 2;  
System.out.println((double)a / b);
```

Output: _____

Q5.

```
byte a = 10, b = 20;  
int c = a + b;  
System.out.println(c);
```

Output: _____

Q6.

```
char ch = 'B';  
System.out.println((int)ch);
```

Output: _____

Q7.

```
int x = 130;  
System.out.println((byte)x);
```

Output: _____

Q8.

```
double d = 25.75;  
System.out.println((int)d);
```

Output: _____

Q9.

```
int x = 65;  
char ch = (char)x;  
System.out.println(ch);
```

Output: _____

Q10.

```
float f = 12.5f;  
double d = f;  
System.out.println(d);
```

Output: _____

Q11.

```
short s = 10;  
int x = s + 5;  
System.out.println(x);
```

Output: _____

Q12.

```
int x = 257;  
byte b = (byte)x;  
System.out.println(b);
```

Output: _____

14. Advanced Challenges

Challenge 1.

```
int a = 5;  
double b = 2;  
double result = a / b + (int)9.99;  
System.out.println(result);
```

Output: _____

Challenge 2.

```
byte x = 10;  
int y = x + 5;  
double z = y / 2;  
System.out.println(z);
```

Output: _____

Challenge 3.

```
double x = 7.8;  
int a = (int)x;  
double b = a / 2;  
System.out.println(a + " " + b);
```

Output: _____

Challenge 4.

```
int x = 5;  
double y = 2.0;  
System.out.println((int)(x / y));
```

Output: _____

Challenge 5.

```
char ch = 'A';  
int x = ch + 2;  
char result = (char)x;  
System.out.println(result);
```

Output: _____

15. Interview Questions

- What is type casting in Java?
- Difference between widening and narrowing casting?
- Why is `int` → `double` automatic but `double` → `int` explicit?
- What happens to the decimal part during `double` → `int` casting?
- What is the output of `(double)(5/2)` and why?
- What is the difference between `(double)a / b` and `(double)(a / b)`?
- Why does `byte + byte` produce an `int`?
- Why can `b += 5` compile for `byte` while `b = b + 5` may not?
- Can `boolean` be cast to `int` in Java?
- What is the difference between primitive casting and reference casting?
- What is `ClassCastException`?
- How can `instanceof` make downcasting safer?

16. 3-Hour Classroom Flow

- 0–15 min: Real-life hook + definition
- 15–40 min: Widening casting
- 40–65 min: Narrowing + data loss
- 65–75 min: Output prediction game
- 75–85 min: Break
- 85–110 min: Integer division traps
- 110–135 min: Promotion + `byte/short/char`
- 135–155 min: Compound assignment
- 155–180 min: Practice
- 180–205 min: Advanced challenges
- 205–225 min: Reference casting intro
- 225–245 min: Interview round
- 245–270 min: Student explain-the-output challenge
- 270–300 min: Quiz + recap + homework

17. One-Page Cheat Sheet

Widening: smaller numeric → larger numeric, usually automatic.

Narrowing: larger numeric → smaller numeric, explicit cast.

double → **int**: fractional part discarded.

int / int: integer division.

(double)int / int: decimal division.

(double)(int / int): too late; decimal information already lost.

byte/short/char arithmetic: commonly promoted to `int`.

boolean: no numeric primitive casting.

Downcasting: explicit and potentially unsafe.